28.5.26

Lecture 5

Lagrange Method

Problem with Vandermond:
1. Computationally expensive for big matrix
2. Not that accurate

Solution: Use lagrange method

$$P_2(x) = l_0(x) f(x_0) + l_1(x) f(x_1) + l_2(x) f(x_2)$$

So,

$$P_n(x) = \sum_{k=0}^{n} l_k(x) f(x_k)$$

($l_k(x)$: Lagrange basis)

Example: $P_3(x) = l_0(x) f(x_0) + l_1(x) f(x_1) + l_2(x) f(x_2) + l_3(x) f(x_3)$

General Form, $l_k(x) = \prod_{\substack{j=0 \\ j \neq k}}^{n} \frac{x - x_j}{x_k - x_j}$

	x	y → f(x)
	age	height
x_0	5	100 — $f(x_0)$
x_1	10	200 — $f(x_1)$
x_2	20	300 — $f(x_2)$

$$P_2(x) = l_0(x) f(x_0) + l_1(x) f(x_1) + l_2(x) f(x_2)$$

① $$l_0(x) = \frac{x - x_1}{x_0 - x_1} \times \frac{x - x_2}{x_0 - x_2} = \frac{x-10}{5-10} \times \frac{x-20}{5-20} = \frac{(x-10)(x-20)}{75}$$

② $$l_1(x) = \frac{x - x_0}{x_1 - x_0} \times \frac{x - x_2}{x_1 - x_2} = \frac{x-5}{10-5} \times \frac{x-20}{10-20} = \frac{(x-5)(x-20)}{-50}$$

③ $$l_2(x) = \frac{x - x_0}{x_2 - x_0} \times \frac{x - x_1}{x_2 - x_1} = \frac{(x-5)(x-10)}{150}$$

Now,

$$P_2(x) = \frac{(x-10)(x-20)}{75} \times 100 + \frac{(x-5)(x-20)}{-50} \times 200 + \frac{(x-5)(x-10)}{150} \times 300$$

Lagrange polynomial expression / Polynomial eqn using Lagrange

$$P_2(x = 30) = 2266.667$$

Drawback: Formula not fixed. Changes if new value is added

$f(x) = \sin x$

	x	$f(x) = \sin x$
x_0	0	0
x_1	$\pi/2$	1
x_2	$\pi/4$	$1/\sqrt{2}$

age	score
25	100
30	150
40	250
50	280
65	290

Use $P_2(x)$ and $P_2(x = 45)$

↳ closet 3টা নিবো cuz P_2. we can only take 3.

$$l_1(x) = \frac{x - x_0}{x_1 - x_0} \times \frac{x - x_2}{x_1 - x_2}$$

Kroneca delta

$$\delta = \begin{cases} 1 ; & i = j \\ 0 ; & i \neq j \end{cases}$$

$l_i(x_j)$

$i = j$: $l_1(x_1) = \frac{x_1 - x_0}{x_1 - x_0} \times \frac{x_1 - x_2}{x_1 - x_2} = 1$

$i \neq j$: $l_1(x_2) = \frac{x_2 - x_0}{x_1 - x_0} \times \frac{x_2 - x_2}{x_1 - x_2} = 0$